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Modified Cramér-von Mises statistics for censored data

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SUMMARY

Cramér–von Mises type statistics are modified so that they can be used to test the goodness of fit of a censored sample, when the distribution function is known completely. Asymptotic percentage points are given, small-sample applications are discussed and asymptotic power considered.

Some key words: Asymptotic distribution; Censored data; Cramér–von Mises statistics; Goodness of fit.

1. INTRODUCTION

Let $x_1, \ldots, x_n$ be independent observations on a random variable X, with a continuous distribution function $G(x)$. If it is wished to test the null hypothesis $H_0: G(x) = G_0(x)$, where $G_0(x)$ is a completely specified distribution function, then this null hypothesis is equivalent to testing that the observations $t_1, \ldots, t_n$, where $t_i = G_0(x_i)$, come from the uniform distribution on $(0, 1)$. If we denote by $F(t)$ the distribution function of $T = G_0(X)$, and $F_n(t)$ the sample distribution function, i.e. the proportion of observations not greater than t, then the Cramér–von Mises type statistics

$$\omega_n^2 = n\int_0^1 \{F_n(t) - t\}^2 \psi(t)\, dt$$

can be used to test the null hypothesis $H_0: F(t) = t$. If $\psi(t) = 1$, ω_n^2 gives the Cramér–von Mises statistic W_n^2, and if $\psi(t) = \{t(1-t)\}^{-1}$, ω_n^2 gives the Anderson–Darling statistic A_n^2. We also consider the Watson statistic U_n^2, introduced as a goodness-of-fit statistic on the circle but which can be used on the line:

$$U_n^2 = n\int_0^1 \left[F_n(t) - t - \int_0^1 \{F_n(s) - s\}\, ds\right]^2 dt.$$

Stephens (1974*a*) and Durbin & Knott (1972) discuss the statistics W_n^2, A_n^2 and U_n^2 in detail and give many references.

In this paper the statistics W_n^2, A_n^2 and U_n^2 are modified so that they can be used to test the goodness of fit of a certain proportion of the random sample, e.g. the sample may be censored. In particular the asymptotic distribution is given for

$$_{q,p}W_n^2 = n\int_q^p \{F_n(t) - t\}^2\, dt, \quad _{q,p}A_n^2 = n\int_q^p \frac{\{F_n(t) - t\}^2}{t(1-t)}\, dt,$$

$$_pU_n^2 = n\int_0^p \left[F_n(t) - t - \frac{1}{p}\int_0^p \{F_n(s) - s\}\, ds\right]^2 dt$$

for various values of p and q $(0 \leqslant q < p \leqslant 1)$. If $q = 0$ then ${}_{q,p}W^2$ is denoted by ${}_pW^2$ and similarly for ${}_pA^2$.

The modification made to U_n^2 is not straightforward. The integral

$$\int_0^1 \{F_n(s) - s\}\, ds$$

can be considered as an estimate of the corresponding integral with $F(t)$, the distribution function of T, replacing $F_n(t)$. If only observations are available which are less than p, then an estimate of this integral is

$$\frac{1}{p}\int_0^p \{F_n(s) - s\}\, ds.$$

2. Asymptotic distribution of the statistics

2·1. *General theory*

Let $Z_n(t) = \sqrt{n}\{F_n(t) - t\}$; then it is well known that $Z_n(t)$ converges weakly to $Z(t)$,where $Z(t)$ is a continuous Gaussian process with mean 0 and covariance,

$$\operatorname{cov}\{Z(t), Z(s)\} = \min(s,t) - st \quad (0 \leqslant s, t \leqslant 1).$$

This means that continuous functionals of $Z_n(t)$ converge in distribution to the same functional of $Z(t)$.

For the asymptotic distribution of the statistics introduced here, we restrict ourselves to considering $Z(t)$ defined in $[q, p]$ rather than $[0, 1]$, and use the theory developed by Durbin (1973, §4·4) for W_n^2 and A_n^2. Thus to find the asymptotic distributions of ${}_{q,p}W_n^2$, ${}_{q,p}A_n^2$ and ${}_pU_n^2$ we need to consider functionals of $Z(t)$. Anderson & Darling (1952) considered W_n^2, which converges in distribution, say to W^2. They showed that

$$W^2 = \int_0^1 Z^2(t)\, dt = \sum_{j=1}^{\infty} C_j \lambda_j^{-1}, \tag{2·1}$$

where the C_j are independent chi-squared random variables with one degree of freedom and the λ_j are eigenvalues of the integral equation

$$\lambda \int_0^1 k(s,t) f(s)\, ds = f(t) \quad (0 \leqslant t \leqslant 1), \tag{2·2}$$

and $k(s,t) = \min(s,t) - st$.

For the asymptotic distributions of ${}_{q,p}W_n^2$ and ${}_{q,p}A_n^2$ we need to consider

$${}_{q,p}W^2 = \int_q^p Z^2(t)\, dt, \quad {}_{q,p}A^2 = \int_q^p Z^2(t)\,(t - t^2)^{-1}\, dt,$$

with $0 \leqslant q < p \leqslant 1$. By the approach of Anderson & Darling (1952) we can show that ${}_{q,p}W^2$ and ${}_{q,p}A^2$ can be expressed as infinite sums of chi-squared random variables. We need to solve the integral equation

$$\lambda \int_q^p k(s,t) f(s)\, ds = f(t) \quad (q \leqslant t \leqslant p), \tag{2·3}$$

for ${}_{q,p}W^2$ and

$$\lambda \int_q^p k(s,t)\{st(1-s)(1-t)\}^{-\frac{1}{2}} f(s)\, ds = f(t) \quad (q \leqslant t \leqslant p), \tag{2·4}$$

for ${}_{q,p}A^2$, where $k(s,t)$ is given above.

For ${}_pU_n^2$ we have to consider the modified process

$$Y_n(t) = Z_n(t) - \frac{1}{p}\int_0^p Z_n(s)\,ds \tag{2·5}$$

for then

$${}_pU_n^2 = \int_0^p Y_n^2(t)\,dt.$$

By noting

$$E\left\{Z_n(t)\int_0^p Z_n(u)\,du\right\} = tp - \tfrac{1}{2}tp^2 - \tfrac{1}{2}t^2, \quad E\left\{\int_0^p Z_n(u)\,du \int_0^p Z_n(v)\,dv\right\} = \tfrac{1}{3}p^3 - \tfrac{1}{4}p^4,$$

we find by expanding $Y_n(t)\,Y_n(s)$ that

$$\begin{aligned}\operatorname{cov}\{Y_n(t), Y_n(s)\} &= k(s,t) - s + \tfrac{1}{2}sp + \tfrac{1}{2}s^2/p - t + \tfrac{1}{2}tp + \tfrac{1}{2}t^2/p + \tfrac{1}{3}p - \tfrac{1}{4}p^2 \\ &= k_1(s,t),\end{aligned} \tag{2·6}$$

say.

The distribution of

$${}_pU^2 = \int_0^p \left\{Z(t) - \frac{1}{p}\int_0^p Z(s)\,ds\right\}^2 dt$$

is given by the sum (2·1), but now the λ_j are eigenvalues of the integral equation

$$\lambda \int_0^p k_1(s,t) f(s)\,ds = f(t) \quad (0 \leqslant t \leqslant p). \tag{2·7}$$

2·2. *Solution of the integral equations*

For the statistic ${}_{q,p}W^2$ the integral equation (2·3) reduces to the differential equation

$$\lambda f(t) + f''(t) = 0 \quad (q \leqslant t \leqslant p)$$

having general solution $f(t) = a\cos(mt) + b\sin(mt)$, where $m = \lambda^{\frac{1}{2}}$. On substituting $f(t)$ into (2·3) it is found that m must satisfy

$$\tan(rm) = m(r-1)/(1 + m^2pq - m^2q), \tag{2·8}$$

where $r = p - q$. Also we find $a/b = \{mq - \tan(mq)\}/\{mq\tan(mq) + 1\}$. With $q = 0$ this gives

$$\tan(pm) = -m(1-p) \tag{2·9}$$

and $f(t) = \surd 2\sin(tm)\{p - \sin(pm)\cos(pm)\,m^{-1}\}^{-\frac{1}{2}}$ is the corresponding normalized eigenfunction. If we put $p = 1$ then we find that (2·9) gives $\lambda_j = j^2\pi^2$ $(j = 1, 2, \ldots)$ in agreement with previous results for W^2.

For the statistic ${}_{q,p}A^2$, we make use of Legendre functions $P_j(x)$. With $q = 0$ and $p = 1$ the eigenvalues of (2·4) are $\lambda_j = j(j+1)$ $(j = 1, 2, \ldots)$ and $f_j(t) = (t - t^2)^{\frac{1}{2}}(d/dt)\,P_j(1-2t)$ $(j = 1, 2, \ldots)$, where

$$P_j(x) = 1 - j(j+1)\,t + (2\,!)^{-2} j(j-1)(j+1)(j+2)\,t^2 - \ldots \quad (-1 \leqslant x \leqslant 1),$$

and $x = 1 - 2t$ (Schelkunoff, 1948, p. 419). For integer values of j, $P_j(1-2t)$ is a jth order polynomial in t. If j is not an integer, we denote j by ν and then $P_\nu(x)$ is an infinite polynomial in x and functionally independent of $P_\nu(-x)$, which we denote by $Q_\nu(x)$.

We find that the eigenfunction to (2·4) is

$$f_\nu(t) = (t-t^2)^{\frac{1}{2}}\left\{a\frac{d}{dt}P_\nu(1-2t)+b\frac{d}{dt}Q_\nu(1-2t)\right\},$$

where

$$a/b = \{Q_\nu(u)(\nu-\nu u+1)+Q_{\nu-1}(u)\}/\{P_{\nu-1}(u)-P_\nu(u)(\nu-\nu u+1)\} \tag{2·10}$$

and ν, a and b satisfy

$$a\{P_\nu(u)(\nu-\nu u+1)+P_\nu(v)(\nu+1+\nu v)-P_{\nu-1}(u)+P_{\nu-1}(v)\} \\ +b\{Q_\nu(u)(\nu-\nu u+1)+Q_\nu(v)(\nu+1+\nu v)-Q_{\nu-1}(v)+Q_{\nu-1}(u)\} = 0, \tag{2·11}$$

where $u = 1-2q$ and $v = 1-2p$. Thus to find $\lambda = \nu(\nu+1)$ we solve (2·11) for ν with a/b given by (2·10).

If $q = 0$, then the solutions are greatly simplified and ν is found by solving

$$(1+\nu v+\nu)P_\nu(v)+P_{\nu-1}(v) = 0$$

with $v = 1-2p$.

For the statistic ${}_pU^2$, the eigenvalues of (2·7) are found by solving

$$\sin(kp) = 0 \tag{2·12}$$

or

$$\tan(kp) = -k(1-p), \tag{2·13}$$

where $k = \frac{1}{2}\lambda^{\frac{1}{2}}$.

The solutions to (2·12) are $k_jp = j\pi$ $(j = 1,2,\ldots)$ or $\lambda_j = 4j^2\pi^2/p^2$, with corresponding eigenfunction, suitably normalized, $f_j(t) = \cos(2kt)$.

The solutions to (2·13) are given by solving (2·9), replacing m by k, and the eigenfunction, suitably normalized, is given by

$$f(t) = \sin(2kt)+k(1-p)\cos(2kt).$$

Thus if we denote the eigenvalues for ${}_pW^2$ by $\{{}_w\lambda_j\}$, that is the solutions of (2·4), and $\{\lambda_j\}$ $\{\lambda_j^*\}$ denote the eigenvalues for ${}_pU^2$, λ_j solutions of (2·12) and λ_j^* solutions of (2·13), then

$$\lambda_j = 4j^2\pi^2/p^2, \quad \lambda_j^* = 4\,{}_w\lambda_j \quad (j = 1,2,\ldots).$$

If $p = 1$, then $\lambda_j = \lambda_j^* = 4j^2\pi^2$ which agrees with Watson's results for U^2.

2·3. *Numerical solution and percentage points*

Eigenvalues for ${}_{q,p}W^2$ and ${}_{q,p}A^2$ were found for $q = 0$ and $p = 0{\cdot}5(0{\cdot}1)0{\cdot}9$, $0{\cdot}95$ and also with $q = 1-p$ for $p = 0{\cdot}75(0{\cdot}05)0{\cdot}95$, and for ${}_pU^2$ with $p = 0{\cdot}5(0{\cdot}1)0{\cdot}9$. This was done by solving the various equations for ${}_{q,p}W^2$, (2·8) and (2·9) using the Newton–Raphson method, and (2·10) and (2·11) with the infinite series for $P_\nu(x)$ truncated to 200 terms.

The percentage points of ${}_{q,p}W^2$, ${}_{q,p}A^2$ and ${}_pU^2$ can be found by using the method of Durbin & Knott (1972). The distribution of the statistic is approximated by B, with

$$B = \sum_{j=1}^{K} C_j\lambda_j^{-1}+aC_v,$$

where the λ_j are the known eigenvalues, solutions of (2·3), (2·4) or (2·7), the C_j are independent chi-squared random variables with one degree of freedom, a and v are constants to be determined, and C_v is a chi-squared random variable with v degrees of freedom. The constants a and v are chosen so that the statistic and B have the same first two moments. Using the twenty smallest eigenvalues the distributions of the statistics can be found accurately by inverting the characteristic function of B, using the method of Imhof (1961). To determine a and v the first two moments of the statistics are needed.

Table 1. *Percentage points of* ${}_pW^2$

Percentage points	Value of p									
	0·50	0·55	0·60	0·65	0·70	0·75	0·80	0·85	0·90	0·95
1	0·0080	0·0095	0·0112	0·0130	0·0149	0·0167	0·0187	0·0206	0·0211	0·0240
2·5	0·0101	0·0120	0·0141	0·0163	0·0186	0·0210	0·0233	0·0256	0·0250	0·0294
5	0·0126	0·0150	0·0176	0·0202	0·0230	0·0258	0·0286	0·0312	0·0314	0·0356
10	0·0166	0·0197	0·0230	0·0264	0·0298	0·0333	0·0367	0·0399	0·0426	0·0450
50	0·0536	0·0619	0·0707	0·0793	0·0877	0·0957	0·1030	0·1093	0·1147	0·1178
90	0·1890	0·2153	0·2407	0·2645	0·2861	0·3048	0·3205	0·3327	0·3412	0·3462
95	0·2579	0·2931	0·3269	0·3581	0·3861	0·4102	0·4298	0·4446	0·4548	0·4599
97·5	0·3295	0·3742	0·4167	0·4558	0·4906	0·5201	0·5439	0·5616	0·5733	0·5791
99	0·4271	0·4847	0·5393	0·5891	0·6330	0·6701	0·6997	0·7212	0·7352	0·7419

Table 2. *Percentage points of* ${}_pA^2$ *and* ${}_pU^2$

Percentage points	(a) Percentage points of ${}_pA^2$ Value of p					(b) Percentage points of ${}_pU^2$ Value of p				
	0·5	0·6	0·7	0·8	0·9	0·5	0·6	0·7	0·8	0·9
1	0·067	0·084	0·103	0·125	0·152	0·0058	0·0083	0·0110	0·0140	0·0171
2·5	0·083	0·104	0·127	0·154	0·186	0·0071	0·0100	0·0133	0·0168	0·0204
5	0·101	0·126	0·154	0·186	0·223	0·0084	0·0118	0·0157	0·0198	0·0239
10	0·128	0·160	0·194	0·234	0·280	0·0104	0·0146	0·0192	0·0241	0·0289
50	0·349	0·425	0·504	0·587	0·676	0·0243	0·0333	0·0428	0·0524	0·0616
90	1·062	1·260	1·451	1·623	1·798	0·0607	0·0804	0·1000	0·1187	0·1360
95	1·419	1·676	1·920	2·146	2·344	0·0778	0·1019	0·1255	0·1476	0·1680
97·5	1·792	2·112	2·421	2·684	2·915	0·0955	0·1241	0·1516	0·1769	0·2000
99	2·301	2·707	3·083	3·419	3·698	0·1197	0·1544	0·1869	0·2162	0·2425

If the general asymptotic form of the Cramér–von Mises statistic is denoted by ω^2, where

$$\omega^2 = \int_q^p X^2(t)\,dt$$

and $X(t)$ is a continuous Gaussian process with symmetric covariance function $\rho(s,t)$, then the mean and variance of ω^2 are given, respectively, by

$$\mu(\omega^2) = \int_q^p \rho(s,s)\,ds, \quad \sigma^2(\omega^2) = 2\int_q^p \int_q^p \rho^2(s,t)\,ds\,dt.$$

These results follow from a straightforward extension of those of Anderson & Darling (1952, §4).

For ${}_{q,p}W^2$, $\rho(s,t) = k(s,t) = \min(s,t) - st$, and so

$$\mu({}_{q,p}W^2) = \tfrac{1}{6}\{p^2(3-2p) - q^2(3-2q)\}$$
$$\sigma^2({}_{q,p}W^2) = \tfrac{4}{3}\{p^4(4 - \tfrac{2}{5}p + \tfrac{1}{6}p^2) + q^3(-\tfrac{1}{6}p^3 + p^2 - p - \tfrac{3}{2}q^2 + \tfrac{3}{4}q)\}.$$

For ${}_{q,p}A^2$, $\rho(s,t) = k(s,t)/\{st(1-s)(1-t)\}^{\frac{1}{2}}$, and so

$$\mu({}_{q,p}A^2) = p-q,$$

$$\sigma^2({}_pA^2) = -4\left\{-\tfrac{1}{2}p^2+2p+(1-p)\log(1-p)+\log(1-p)\log(p)-\tfrac{1}{6}\pi^2+(1-p)\right.$$
$$\left.+\frac{(1-p)^2}{2^2}+\frac{(1-p)^3}{3^2}+\ldots\right\}.$$

The last terms in $\sigma^2({}_pA^2)$ are the integral

$$\int_0^p \log(1-t)\,t^{-1}dt,$$

Table 3. *Percentage points of ${}_{1-p,\,p}W^2$ and ${}_{1-p,\,p}A^2$ for symmetric double censoring*

(a) Percentage points of ${}_{1-p,p}W^2$ (b) Percentage points of ${}_{1-p,p}A^2$

Percentage point	(a) Value of p: 0·75	0·80	0·85	0·90	0·95	(b) Value of p: 0·75	0·80	0·85	0·90	0·95
1	0·0083	0·0113	0·0143	0·0177	0·0221	0·043	0·061	0·083	0·110	0·145
2·5	0·0112	0·0149	0·0188	0·0229	0·0278	0·055	0·078	0·105	0·137	0·179
5	0·0147	0·0192	0·0239	0·0288	0·0342	0·071	0·098	0·130	0·168	0·216
10	0·0202	0·0260	0·0320	0·0381	0·0438	0·095	0·129	0·169	0·216	0·271
50	0·0718	0·0862	0·0993	0·1102	0·1166	0·321	0·402	0·488	0·578	0·674
90	0·2621	0·2956	0·3206	0·3358	0·3445	1·134	1·322	1·498	1·661	1·810
95	0·3592	0·4018	0·4311	0·4484	0·4584	1·546	1·784	2·000	2·194	2·362
97·5	0·4612	0·5131	0·5462	0·5661	0·5775	1·976	2·266	2·525	2·752	2·940
99	0·6023	0·6665	0·7030	0·7269	0·7401	2·562	2·925	3·243	3·513	3·730

which is related to the Debye function (Abramowitz & Stegun, 1964, §27.1.1). The function is tabulated but the tables would need interpolating for the values of $y = -\log(1-p)$, so that it is easier to compute the above integral by truncating the series. Also we have

$$\sigma^2({}_{q,p}A^2) = \sigma^2({}_pA^2)-\sigma^2({}_qA^2)+4\{q+\log(1-q)\}\{\log(p/q)-p+q\}.$$

The moments of ${}_pU^2$ are found from ${}_pW^2$ and W^2. Using the relationship between the eigenvalues for ${}_pU^2$ and for ${}_pW^2$, it can be shown that

$$\mu({}_pU^2) = \tfrac{1}{4}\mu({}_pW^2)+\tfrac{1}{4}p^2\mu(W^2), \quad \sigma^2({}_pU^2) = \tfrac{1}{16}\sigma^2({}_pW^2)+\tfrac{1}{16}p^4\sigma^2(W^2).$$

This follows from the representation (2·1) of the statistics. The percentage points of the statistics are given in Tables 1, 2 and 3.

3. Small-sample applications and power considerations

We can find expressions for the statistics involving the observations $t_1, \ldots, t_n$ explicitly. If $t_{(1)} < t_{(2)} < \ldots < t_{(R)} \leqslant p < t_{(R+1)} < \ldots t_{(n)}$, that is R observations are less than p and $t_{(R+1)}, \ldots, t_{(n)}$ are censored, then

$${}_pW_n^2 = \sum_{i=1}^{R}\left(t_{(i)}-\frac{2i-1}{2n}\right)^2-\frac{R(4R^2-1)}{12n^2}+np\left(\frac{R^2}{n^2}-p\frac{R}{n}+\tfrac{1}{3}p^2\right),$$

and obviously ${}_{q,p}W_n^2 = {}_pW_n^2-{}_qW_n^2$. Also

$${}_pA_n^2 = \sum_{i=1}^{R}\{(2i-1)/n\}\{\log(1-t_{(i)})-\log t_{(i)}\}-2\sum_{i=1}^{R}\log(1-t_{(i)})$$
$$+n\left\{\frac{2R}{n}-\left(\frac{R}{n}\right)^2-1\right\}\log(1-p)+\frac{R^2}{n}\log p-pn,$$

and similarly ${}_{q,p}A_n^2 = {}_pA_n^2 - {}_qA_n^2$. Finally ${}_pU_n^2$ is given by

$${}_pU_n^2 = {}_pW_n^2 - np\left\{\frac{R}{n} - \tfrac{1}{2}p - \frac{1}{pn}\sum_{i=1}^{R} t_{(i)}\right\}^2.$$

An approximation was made to the distribution of ${}_pW_n^2$ using the first three moments of ${}_pW_n^2$, and with evidence from Monte Carlo experiments for ${}_pA_n^2$, the distributions of ${}_pW_n^2$ and ${}_pA_n^2$ converge quickly to their asymptotic distributions. For $p = 0{\cdot}5$ and $n = 10$ there is a difference only in the third significant figure of the percentage points of ${}_pW^2$ and ${}_pW_n^2$. For $p = 0{\cdot}75$ and $0{\cdot}9$ there is a difference of one digit in the second significant figure for $n = 30$.

The statistic

$${}_rW_n^2 = n\int_0^{t_{(r)}} \{F_n(t) - t\}^2\,dt$$

also converges to ${}_pW^2$ provided $r/n \to p$ as $n \to \infty$, where $t_{(r)}$ is the rth order statistic of the observations $t_1, \ldots, t_n$. Similarly, the statistic ${}_rA_n^2$ can be defined and ${}_rA_n^2$ converges to ${}_pA^2$. The small-sample distribution of ${}_rW_n^2$ was investigated, using a three-moment fit to the distribution of ${}_rW_n^2$ and supporting Monte Carlo distributions. It was found that the distribution of ${}_rW_n^2$ is slow to converge to ${}_pW^2$ in the upper tail, and for significance points of ${}_rW_n^2$ reference should be made to the first author's unpublished Nottingham thesis. The expression for ${}_rW_n^2$ is given by replacing p by $t_{(r)}$ and R by $(r-1)$ in the expression for ${}_pW_n^2$; similar comments hold for ${}_rA_n^2$.

A power study of the statistics ${}_pW^2$ and ${}_pA^2$ was made similar to that of Durbin & Knott (1972) for W^2, A^2 and U^2, for shifts in mean or variance from a null hypothesis specifying a normal distribution with mean and variance known. Comparisons can be made with their results in their Tables 3 and 4. It was found for mean shift ${}_pW^2$, where $p = 0{\cdot}9$, has power which differs from that of W^2 in the third significant digit only, and ${}_pW^2$, where $p = 0{\cdot}5$, has power about 10% less than W^2, for the same alternatives and significance level, considered by Durbin & Knott. Similar remarks hold for ${}_pA^2$ compared with A^2 for mean shift. Here W^2 and A^2 have little power for variance shift and similarly ${}_pW^2$ and ${}_pA^2$ have little power; see Stephens (1974*b*) for comments on the power of A^2 for variance shift. It would be expected that ${}_pU^2$ has power against variance shift, as does U^2.

Kolmogorov–Smirnov type statistics for this problem have been considered. Rényi (1953) and Birnbaum & Lientz (1969) have considered one-sided statistics, and Barr & Davidson (1973) tabulated the distribution of the two-sided Kolmogorov–Smirnov statistic.

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